RESEARCH PROJECT & PROPOSAL APPROVAL BRIEF

Cascaded Adaptive Blended RAG: Optimizing Retrieval Efficiency via Pre-Retrieval Query Cascading

Author: Master's Student | **Target Base Paper:** Sawarkar et al. (2024) | **Focus:** RAG Efficiency & Adaptive Routing

1. Base Paper Selection & Motivation

Selected Base Paper: *'Blended RAG: Improving RAG (Retriever-Augmented Generation) Accuracy with Semantic Search and Hybrid Query-Based Retrievers'* (Sawarkar et al., 2024).

Why This Paper? The overall accuracy of Retrieval-Augmented Generation (RAG) systems heavily depends on retriever quality. Sawarkar et al. demonstrated that combining **dense vector indexes** and **sparse encoder indexes** with **multi-field hybrid query strategies** (such as *Best Fields* search across titles, abstracts, and body contexts) significantly outperforms single-retriever baselines (e.g., standard dense embeddings or BM25 keyword search) and avoids costly domain-specific model fine-tuning.

The Core Problem & Research Gap: While Blended RAG achieves top-tier retrieval accuracy across benchmarks like NQ and HotpotQA, it suffers from a major computational bottleneck: **The Static Hybrid Penalty**. Blended RAG executes full multi-field hybrid queries (Dense Vector Search + Sparse ELSER Search + Multi-Field Expansion) for *every incoming query regardless of difficulty*. Simple factual queries that could easily be answered by a fast lexical lookup pay the exact same latency and compute cost as complex compositional queries.

2. Proposed Research Extension: Cascaded Adaptive Blended RAG

> **Core Objective:** Develop a lightweight **Pre-Retrieval Cascaded Router** that dynamically evaluates query difficulty before search execution. Simple queries exit early via a fast BM25 pass, while difficult queries escalate to full Blended RAG (BM25 + SPLADE + Dense MiniLM). **Goal:** Reduce retrieval latency and vector search calls by 40–60% while maintaining 95%+ of full Blended RAG accuracy.

Key Architectural Components:

- **Tier 0 (Pre-Retrieval Analysis):** Extracts fast, zero-overhead features from the query text alone (length, rare token count, IDF score statistics, entropy).
- **Tier 1 (BM25 Fast Pass & Early Exit):** Runs a near-zero-cost lexical search. If the top-1 score s_1 exceeds a threshold τ_{high} or the confidence margin $\Delta = s_1 - s_2$ is wide, the system returns the BM25 candidates immediately and skips all vector search.
- **Tier 2 (Fallback Blended Search):** If BM25 confidence is low ($\Delta < \tau_{low}$), the query escalates to a 3-way hybrid search: **BM25 + SPLADE (Sparse Learned) + Dense MiniLM** merged via Reciprocal Rank Fusion (RRF).
- **Metadata-Aware Query Expansion:** Evaluates multi-field matching (*Best Fields* across title, abstract, and content) to preserve Blended RAG's core innovation on structured datasets.

Distinguishing from Recent Literature (2025–2026):

Recent papers like *DAT* (Hsu et al., 2025) and *QuDAR* (Kim et al., 2026) focus on dynamic per-query alpha weighting (α-tuning). However, they still execute both sparse and dense retrievers for 100% of queries, saving no execution latency. In contrast, our proposed **Cascaded Architecture** enforces early termination, providing measurable CPU/compute savings.

3. Methodology & System Architecture

The system pipeline processes incoming queries sequentially through early-exit logic, as illustrated in the framework mapping below:

Pipeline Stage	Mechanism / Models Used	Compute Cost	Target Query Type
Tier 0: Pre-Retrieval	Query Length, IDF Statistics, Rare Tokens	< 1 ms (CPU)	All incoming user queries
Tier 1: BM25 Fast Pass	BM25 Lexical Index + Margin Check (Δ)	~ 2-5 ms (CPU)	Simple factual / exact-keyword match
Tier 2: Blended Hybrid	BM25 + SPLADE Sparse + Dense MiniLM + RRF	~ 30-80 ms (CPU)	Complex, compositional, or abstract
Field Strategy	Best Fields / Cross Fields on Title & Abstract	Zero added index cost	Metadata-rich corpora (e.g. SciFact)

4. Experimental Setup & Evaluation Protocol

To rigorously validate the extension, experiments will be conducted on CPU-constrained laptop environments using open-source benchmarks from the BEIR suite:

- **Primary Datasets:** *BEIR SciFact* (5,183 scientific abstracts with title/abstract metadata) and *Natural Questions (NQ)*.
- **Baselines to Compare:** (1) Standard BM25, (2) Dense MiniLM Vector Search, (3) Static Blended RAG (Sawarkar et al.), (4) Cascaded Adaptive Blended RAG (Proposed).

Evaluation Dimensions & Metrics:

Dimension	Specific Metrics	Target Outcome / Benchmark Goal
Retrieval Quality	Recall@10, MRR@10, nDCG@10	Maintain >= 95% of static Blended RAG accuracy
Computational Efficiency	Latency (ms/query), % Vector Calls Avoided	Reduce mean retrieval latency by 40-60%
Downstream Generation	Exact Match (EM) & F1 Score (via Llama-3.2-3B)	Ensure zero degradation in generator answer factual accuracy

5. Implementation Roadmap & Milestones

Phase	Milestone Task	Key Deliverable
Phase 1 (Week 1)	Setup BEIR SciFact index; build BM25 & Dense baselines	Baseline evaluation scripts & benchmark logs
Phase 2 (Week 2)	Implement Static Blended RAG with BM25 + SPLADE + RRF	Reproduction of multi-field Blended RAG accuracy
Phase 3 (Week 3)	Develop Tier 0 query feature extractor & Tier 1 cascade router	Tuned threshold protocol (\(\tau_{high}, \Delta\))
Phase 4 (Week 4)	End-to-end evaluation (Retrieval + 3B Local LLM Generation)	Final Thesis Chapter & Latency vs. Accuracy Trade-off Plots

Summary for Advisor Approval: This research addresses an unresolved efficiency bottleneck in state-of-the-art hybrid RAG systems. It is technically grounded, reproducible on accessible hardware, and builds directly upon foundational literature (Sawarkar et al., 2024). Approval is requested to proceed with Phase 1 benchmarking.